MATH 152
Mrs. Bonny Tighe

PART TWO
EXAM IIA
50 points
7.4-11.3

NAME Solutions

TA_____ Wed. 10/25/2017

1. (10 points) Are the following improper integrals convergent or divergent? Describe what convergence means.

a) $\int_0^3 \frac{4x}{\sqrt{9-x^2}}\,dx$

$$\lim_{t\to 3^+} \int_0^t \frac{4x}{\sqrt{9-x^2}}\,dx$$

$u = 9-x^2$
$du = -2x\,dx$

$$\lim_{t\to 3^+} -2\int u^{-1/2}\,du$$

$$-2\,\frac{1}{1/2}\,u^{1/2}$$

$$\lim_{t\to 3^+} \left[-4\sqrt{9-x^2}\right]_0^t$$

$$\lim_{t\to 3^+} \left[\left(-4\sqrt{9-t^2}\right)-\left(-4\sqrt{9}\right)\right]$$

$$\lim_{t\to 3^+} \left[-4(0)+4(3)\right] = 12$$

so convergent

b) $\int_1^\infty x^2 e^{-x^3}\,dx,$

$$\lim_{t\to\infty} \int_1^t e^{-x^3} x^2\,dx$$

$u = -x^3$
$du = -3x^2\,dx$

$$\lim_{t\to\infty} -\frac{1}{3} e^{-x^3}\Big]_1^t$$

$$\lim_{t\to\infty} \left[-\frac{1}{3} e^{-t^3} + \frac{1}{3} e^{-1}\right]$$

$$\lim_{t\to\infty} \left[-\frac{1}{3e^{t^3}} + \frac{1}{3e}\right] =$$

$1/3e$ so

congruent

2. (10 points) Find the length of the curve $x = \ln(y^2 - 1)$ $2 \le y \le 3$

dy

$$dx/dy = \frac{2y}{y^2-1} \quad 1$$

$$L = \int_a^b \sqrt{1 + (dx/dy)^2}\, dy \quad 1$$

$$L = \int_2^3 \sqrt{1 + \frac{4y^2}{(y^2-1)^2}}\, dy = \int_2^3 \sqrt{\frac{y^4 - 2y^2 + 1 + 4y^2}{(y^2-1)^2}}\, dy \quad 1$$

$$L = \int_2^3 \frac{\sqrt{(y^2+1)^2}}{y^2-1}\, dy = \int_2^3 \frac{y^2+1}{y^2-1}\, dy$$

$$\begin{array}{r} 1 \\ y^2-1 \overline{\big) y^2+1} \\ -(y^2-1) \\ \hline 2 \end{array}$$

$$\int_2^3 1 + \frac{2}{(y+1)(y-1)}\, dy$$

$$\frac{A}{y+1} + \frac{B}{y-1} = \frac{2}{(y+1)(y-1)}$$

$$A(y-1) + B(y+1) = 2$$

$y = 1$ $\quad B = 1$

$y = -1$ $\quad A = -1$

$$\int_2^3 \left(1 + \frac{1}{y-1} - \frac{1}{y+1}\right) dy$$

$$y + \ln\left|\frac{y-1}{y+1}\right| \Big]_2^3$$

$$\left(3 + \ln\left|\frac{2}{4}\right|\right) - \left(2 + \ln\left|\frac{1}{3}\right|\right)$$

$$1 + \ln\left(\frac{1}{2}\right) - \ln(1/3) = 1 + \ln(3/2)$$

$y^{2/3} = 9 - x^{2/3}$

$y = (9 - x^{2/3})^{3/2}$

3. (15 points) a) The given curve is rotated about the ~~y~~ x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius \;\; arclength$ $\quad y^{2/3} + x^{2/3} = 9, \;\; 1 \le x \le 8$

$(r = y)$ 2

$dy/dx = \frac{3}{2}(9 - x^{2/3})^{1/2}(-\frac{2}{3}x^{-1/3})$

$= -\frac{(9 - x^{2/3})^{1/2}}{x^{1/3}}$

$SA = 2\pi \int_1^8 y\sqrt{1 + (dy/dx)^2}\, dx$

$SA = 2\pi \int_1^8 (9 - x^{2/3})^{3/2}\sqrt{1 + \frac{9 - x^{2/3}}{x^{2/3}}}\, dx = 2\pi \int_1^8 (9 - x^{2/3})^{3/2}\sqrt{\frac{x^{2/3} + 9 - x^{2/3}}{x^{2/3}}}\, dx$

$2\pi \int_1^8 (9 - x^{2/3})^{3/2} \frac{3}{x^{1/3}}\, dx$

$u = 9 - x^{2/3}$

$du = -\frac{2}{3}x^{-1/3}\, dx$

$-\frac{3}{2}x^{1/3} = dx$

$2\pi \int u^{3/2} \frac{3}{x^{1/3}}\left(-\frac{3}{2}x^{1/3}\right) du$

$2\pi \int -\frac{9}{2} u^{3/2}\, du = -9\pi \left[\frac{1}{5/2}(9 - x^{2/3})^{5/2}\right]_1^8 = -9\pi\left(\frac{2}{5}\right)(9 - x^{2/3})^{5/2}\Big]_1^8$

$-\frac{18\pi}{5}\left[(9 - 8^{2/3})^{5/2} - (9 - 1)^{5/2}\right]$

4. (10 points) a) How large should we take n in order to guarantee that the Midpoint approximation for $\int_2^4 (\ln x)dx$ is accurate to within 0.01? (see next page)

$f(x) = \ln x$

$f'(x) = x^{-1}$

$f''(x) = -x^{-2} = -\frac{1}{x^2}$

$K = \left|\frac{1}{x^2}\right| \quad 2 \le x \le 4$

$K = 1/4$

$|E_M| \le \frac{\frac{1}{4}(4-2)^3}{24n^2} \le \frac{1}{10^2}$

$\frac{1}{12n^2} \le \frac{1}{10^2}$

$10^2 \le 12n^2$

$\sqrt{\frac{100}{12}} \le n$

next integer

b) Estimate the error in using the partial sum S_{10} of the series $\sum_{k=1}^{\infty} \frac{1}{k^4}$

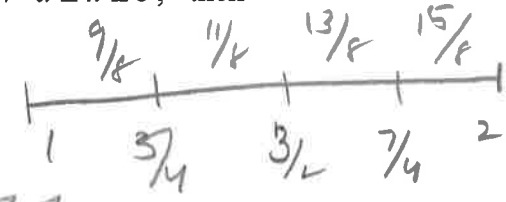
$\lim_{t \to \infty} \int_{10}^{t} k^{-4}\, dk = \lim_{t \to \infty} \left[\frac{1}{-3}k^{-3}\right]_{10}^{t} =$

$\lim_{t \to \infty} \left[-\frac{1}{3t^3} + \frac{1}{3(10)^3}\right] = \frac{1}{3(10)^3}$

$\frac{1}{3(11)^3} \le R_{10} \le \frac{1}{3(10)^3}$

5. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

$$|E_M| \le \frac{K(b-a)^3}{24n^2} \qquad \int_1^2 \frac{1}{e^x}dx, \quad n = 4$$



$\Delta x = \frac{1}{4}$

a) $A_M = \frac{1}{4}\left[e^{-9/8} + e^{-11/8} + e^{-13/8} + e^{-15/8}\right]$

b) $A_T = \frac{1}{8}\left[e^{-1} + 2e^{-5/4} + 2e^{-3/2} + 2e^{-7/4} + e^{-2}\right]$

c) $A_S = \frac{1}{12}\left[e^{-1} + 4e^{-5/4} + 2e^{-3/2} + 4e^{-7/4} + e^{-2}\right]$

$f(x) = e^{-x}$

$f'(x) = -e^{-x}$

$f''(x) = e^{-x} = \frac{1}{e^x}$

on $1 \le x \le 2$

$K = \frac{1}{e}$

$|E_M| \le \frac{\frac{1}{e}(2-1)^3}{24(4)^2}$

$|E_M| \le \frac{1}{e(24)(16)}$

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a) $\int_0^2 \frac{3x}{\sqrt{4-x^2}}\,dx$

$\lim_{t\to 2^-} \int_0^t \frac{3x}{\sqrt{4-x^2}}\,dx$

$u = 4 - x^2$

$du = -2x\,dx$

$-\frac{3}{2}\int u^{-1/2}\,du =$

$-3/2 \; 1/1/2\, u^{1/2}$ $\quad -3/2 \cdot \frac{2}{1}$

$\lim_{t\to 2^-} -3\sqrt{4-x^2}\Big]_0^t$

$\lim_{t\to 2^-} \left[-3\sqrt{4-t^2} + 3\sqrt{4}\right]$

$-3\sqrt{0} + 3\sqrt{4}$

6 So convergent

b) $\int_1^\infty x^3 e^{-x^4}\,dx,$

$\lim_{t\to\infty} \int_1^t e^{-x^4} x^3\,dx$

$u = -x^4$

$du = -4x^3\,dx$

$\lim_{t\to\infty} \int_1^t -\frac{1}{4} e^u\,du =$

$\lim_{t\to\infty} -\frac{1}{4} e^{-x^4}\Big]_1^t =$

$\lim_{t\to\infty} -\frac{1}{4e^{x^4}}\Big]_1^t =$

$\lim_{t\to\infty} \left[-\frac{1}{4e^{t^4}} + \frac{1}{4e^1}\right] = \frac{1}{4e}$

So convergent

2. (10 points) Find the length of the curve $y = \ln(x^2 - 1)$ $2 \le x \le 3$

$$L = \int_a^b \sqrt{1 + (dy/dx)^2}\, dx \qquad dy/dx = \frac{2x}{x^2-1}$$

$$L = \int_2^3 \sqrt{1 + \frac{4x^2}{(x^2-1)^2}}\, dx =$$

$$L = \int_2^3 \sqrt{\frac{(x^2-1)^2 + 4x^2}{(x^2-1)^2}}\, dx = \int_2^3 \frac{\sqrt{x^4 - 2x^2 + 1 + 4x^2}}{x^2-1}\, dx$$

$$L = \int_2^3 \frac{x^2+1}{x^2-1}\, dx = \int_2^3 1 + \frac{2}{x^2-1}\, dx$$

$$\begin{array}{r} 1 \\ x^2-1 \overline{\big) x^2+1} \\ -(x^2-1) \\ \hline 2 \end{array}$$

$$\frac{A}{x+1} + \frac{B}{x-1} = \frac{2}{x^2-1}$$

$A(x-1) + B(x+1) = 2$

$x = 1 \qquad B = 1$

$x = -1 \qquad A = -1$

$$\int_2^3 \left(1 + \frac{1}{x-1} - \frac{1}{x+1}\right) dx$$

$$x + \ln\left|\frac{x-1}{x+1}\right| \Big]_2^3 = \left(3 + \ln\left(\tfrac{2}{4}\right)\right) - \left(2 + \ln\left(\tfrac{1}{3}\right)\right)$$

$$1 + \ln(1/2) - \ln(1/3) \quad \text{or}$$

$$1 + \ln(3/2)$$

$y^{2/3} = 4 - x^{2/3} \qquad y = (4 - x^{2/3})^{3/2}$

3. (15 points) a) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b \textit{radius arclength}$ $\qquad y^{2/3} + x^{2/3} = 4, \quad 1 \le x \le 2$

dx

$$\frac{dy}{dx} = \frac{3}{2}(4 - x^{2/3})^{1/2}\left(-\frac{2}{3}x^{-1/3}\right) = \frac{-\sqrt{4 - x^{2/3}}}{x^{1/3}}$$

$$S = 2\pi \int_1^2 y\sqrt{1 + \frac{4 - x^{2/3}}{x^{2/3}}}\,dx = 2\pi \int_1^2 (4 - x^{2/3})^{3/2}\sqrt{\frac{x^{2/3} + 4 - x^{2/3}}{x^{2/3}}}$$

$$S = 2\pi \int_1^2 (4 - x^{2/3})^{3/2}\frac{2}{x^{1/3}}\,dx \qquad u = 4 - x^{2/3}, \quad du = -\tfrac{2}{3}x^{-1/3}\,dx = dx$$

$$-\frac{3}{2}x^{1/3}$$

$$S = 2\pi \int_1^2 u^{3/2}\frac{2}{x^{1/3}}\left(-\frac{3}{2}x^{1/3}\right)dx$$

$$-6\pi \cdot \frac{1}{5/2}u^{5/2}\Big]_1^2 = -6\pi\left[\frac{2}{5}(4 - x^{2/3})^{5/2}\right]_1^2 =$$

$$-\frac{12\pi}{5}\left[(4 - 2^{2/3})^{5/2} - (4 - 1^{2/3})^{5/2}\right]$$

4. (10 points) a) How large should we take n in order to guarantee that the Midpoint approximation for $\int_1^2 (\ln x)dx$ is accurate to within 0.001? (see next page)

$f(x) = \frac{1}{x} = x^{-1}$ $\qquad k = 1 \qquad |E_n| \le \frac{1(2-1)^3}{24n^2} \le \frac{1}{10^3}$

$f'(x) = -x^{-2} = -\frac{1}{x^2}$ $\qquad 10^3 \le 24n^2$

on $1 \le x \le 2$ $\qquad \sqrt{\frac{10^3}{24}} \le n$ the next integer

largest at $x = 1$

b) Estimate the error in using the partial sum S_{20} of the series $\sum_{k=1}^{\infty}\frac{1}{k^3}$

$$\lim_{t\to\infty}\int_{20}^{t} k^{-3}\,dk = \lim_{t\to\infty}\frac{1}{-2}k^{-2}\Big]_{20}^{t} = \lim_{t\to\infty} -\frac{1}{2k^2}\Big]_{20}^{t}$$

$$\lim_{t\to\infty}\left[-\frac{1}{2t^2} + \frac{1}{2(20)^2}\right] = \frac{1}{2(20)^2}$$

$$\frac{1}{2(21)^2} \le R_{20} \le \frac{1}{2(20)^2}$$

5. (10 points) Use (a) the Midpoint Rule (b) the Trapezoidal Rule and (c) Simpson's Rule to approximate the given integral with the specified value of n. Find the error bounds for the Midpoint rule. Suppose $|f''(x)| \le K$ *for* $a \le x \le b$, then

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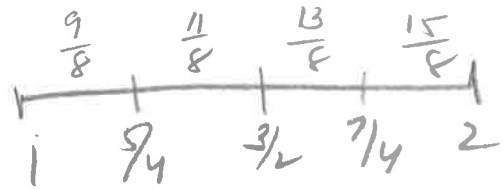
$\Delta x = 1/4$



a) $A_M = \frac{1}{4}\left[\sqrt{9/8} + \sqrt{11/8} + \sqrt{13/8} + \sqrt{15/8}\right]$

b) $A_T = \frac{1}{8}\left[\sqrt{1} + 2\sqrt{5/4} + 2\sqrt{3/2} + 2\sqrt{7/4} + \sqrt{2}\right]$

c) $A_S = \frac{1}{12}\left[\sqrt{1} + 4\sqrt{5/4} + 2\sqrt{3/2} + 4\sqrt{7/4} + \sqrt{2}\right]$

$f(x) = x^{1/2}$

$f'(x) = \frac{1}{2}x^{-1/2}$

$f''(x) = -\frac{1}{4}x^{-3/2} = \left|\frac{1}{4x^{3/2}}\right|$

on $1 \le x \le 2$

$K = \frac{1}{4}$

$|E_M| \le \frac{1/4\,(2-1)^3}{24(4)^2}$

$|E_M| \le \frac{1}{4(24)(16)}$